

24 March 2025

Enhancing Credit Decisions with AI: A Pipeline for Automated Financial Document Analysis and Creditworthiness Prediction

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Keywords

computing and processing

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March 18, 2025

ABSTRACT

The transformational potential of AI-driven credit analysis for business-to-business transactions is examined in this paper. The Financial Document Scanner, Credit Limit Recommendation, and Order Release Recommendation are the three main use cases discussed in this paper. The introduction provides a strong basis for understanding business-to-business (B2B) transactions and the crucial function that credit analysis plays in the Order-to-Cash (O2C) cycle. This paper describes the procedure for gathering, preparing, and analyzing of data for each use case in the methodology section. A multi-step procedure for the Financial Document Scanner to extract relevant financial metrics from the financial accounts of various firms is employed to guarantee precise data extraction, this includes image improvement, noise reduction, and optical character recognition (OCR). On the other hand, a large dataset that includes invoice data, past payment history, credit history, and external market intelligence is utilized in the Credit Limit Recommendation use case. The methodology described in this paper leverages sophisticated analytical tools and different datasets to produce rapid, accurate, and actionable credit risk evaluations. Artificial Intelligence (AI)-driven B2B credit analysis provides unmatched chances to optimize credit management strategies, streamline operations, and promote sustainable growth in today's dynamic business environment. These opportunities are achieved through automating financial document processing, improving order block prediction, and utilizing domain-specific insights.

1 Introduction

In today's changing business world where companies deal with other companies, the integration of Artificial Intelligence (AI) integration is causing a revolution in conventional practices. After looking at different approaches [1], [2], [3] etc., A significant technological gap in B2B credit analysis is noticed. There has been no single paper on end to end to automated B2B credit analysis which helps in reducing the customers credit analysis time significantly. This paper looks at how AI helps with B2B credit assessments especially when it comes to automating the the analysis of financial statements and predict if order should get stopped. Before getting into cutting edge approach, This section explains the basics of B2B transactions, order-to-cash process, and critical role of credit analysis within this framework.

1.1 Understanding B2B Transactions

Business-to-business (B2B) deals form the backbone of the world economy covering a broad spectrum of company-to-company interactions [4]. B2B transactions happen when two or more businesses trade with each other, unlike business-to-consumer (B2C) deals where individual buyers get products or services from companies [4]. These exchanges might include distribution agreements or procurement of raw materials and outsourcing work. B2B transactions are primarily distinguished by transactions scale, level of complexity, and the importance of long-term connections and collaborations.

The fundamental goal of B2B transactions is win-win situations where both parties seek to maximize value while minimizing risks [3]. This often means having open discussions, carrying out agreements, and giving credit in order to facilitate transactions. Credit risks must be appropriately managed and tracked due to the volume and complexity of business-to-business transactions. Over time, this maintains businesses operating and profitable.

1.2 The Order-to-Cash (O2C) Cycle

The O2C cycle helps in defining the inception and culmination of every transaction from the moment a customer/business places an order to the time money finally reaches the seller's pocket [5]. It involves a number of interlinked processes that start with order management, followed by invoicing, delivery, and finally, payment collection. Among these, credit analysis comes to the forefront in terms of deciding on credit to customers and associated risks.

1.3 Credit Analysis in the O2C Cycle

Credit analysis is at a crucial place in the O2C cycle because it is the point where all the possible risks can converge [5]. It involves the scrutiny of financial statements and other indicators of customers or the business partners willing to take credit and other similar facilities. The most important financial statements such as balance sheets, income statements, and the statement of cash flows which are used primarily in the process of credit analysis of a company.

The main goals of the credit analysis are twofold: to assess whether a customer will be in a position to pay or not, and to determine appropriate credit terms in view of risk-reward [6]. In order to project such views concerning the financial health and stability of a business entity, the credit analysts look forward to studying financial metrics on liquidity, solvency, profitability, and cash flow adequacy. Based on such assessments, credit decisions on different levels are made regarding the amount of credit to be extended, duration of the credit terms, and requirements of additional collateral.

Traditionally, credit risk analysis has been resource-intensive and generally relied on some kind of manual review with subjective judgment. The analysts seem to wade through thick sets of financial documents, compute ratios, enforce industry comparisons, and finally arrive at the credit decisions. While this approach yields reasonably accurate results, it inherently is time consuming, resource intensive, and prone to human biases.

1.4 Impact of Generic vs Domain-Specific Credit Analysis

A broad, uniform framework is used in generic credit analysis to assess the relative creditworthiness of companies in a wide range of industries. It is predicated on widely used financial measures, such as debt-to-equity ratios, historical payment patterns, and liquidity ratios, all of which are applicable in all situations and with the assurance of a uniform set of criteria of generic credit analysis, this would foster consistency and comparability in the credit evaluation so that would be easier to benchmark and assess credit risk across a diversified portfolio of clients. This method has the benefits of simplicity and scalability since it can be used to a wide range of industries with minimal adjustment. Furthermore, AI-driven systems are good at doing generic credit analysis by automating the extraction of standard financial metrics from various datasets and interpreting the same. This greatly reduces the time and effort needed to carry out credit assessments, improving efficiency and accurate results in the process. Generic credit analysis, in particular, benefits organizations dealing with a variety of clients across sectors through provision of a reliable, homogeneous method for assessing credit risk. On the contrary, domain-specific credit analysis is tailored to the industry's peculiarities and specific needs. It considers industry-specific factors relating to regulatory compliance, market dynamics, operational risks, and industry benchmarks. For instance, while credit risk in a manufacturing company is based on supply chain stability, inventory turnover, and production capacity, a technology company would base its credit risk assessment on its innovation potential through intellectual property and market penetration. The credit scoring models, which are domain-specific in nature, can provide a better and more comprehensive context to the creditworthiness of a person, allowing insights that otherwise may have been left undetected by the generic ways and means. Since this requires an in-depth understanding of the concerned industry, the input of specialized skills and knowledge is usually integrated into the analysis; AI models can be trained to identify and focus on domain-specific factors. This is particularly useful for sectors that have special risk profiles or are tightly governed by regulations.

In financial document processing, a core aspect of credit analysis, the traditional approach has been manual extraction, interpretation, and analysis of financial information derived from several sources. Apart from being time consuming task, it is also riddled with inaccuracies and inconsistencies due to human limitations. AI-driven solutions offer an interesting alternative by way of automating the extraction, normalization, and analysis of financial information from unstructured documents such as financial statements, invoices, and credit reports.

AI systems will have more efficient machine learning algorithms and OCR techniques that can run through the enormous amounts of textual and numerical information at faster and more accurate speeds than ever before. Such systems are

capable of processing the identification of key data points, recognizing abnormal patterns in those data, and providing red flags for unusual cases for further scrutiny. Further, AI algorithms will learn and adapt to market conditions, respond continuously to changing regulatory requirements and customer behaviors, thereby making credit analysis processes even more agile and responsive.

One of the most important applications of AI in B2B credit analysis is order block prediction, wherein AI algorithms analyze transactional data to identify a potential order block or delay in payment. Such AI systems are capable of forecasting cash flow disruptions and thereby proactively mitigating risks by analyzing past transactional patterns, customer behavior, and market dynamics. This proactive stance will enable companies to set up focused interventions, by line of credit limit reviews, issuance of payment reminders, or renegotiation of terms, with a view to preventing order blocks and optimizing cash flows.

The transformation of the B2B credit analysis process, among other areas, opens the door to wide organizational effectiveness and possibly competitive strategic value. The utilization of AI technologies to automate repetitively driven tasks, support decision-making processes in the form of decision support systems, and enable learning from sources of vast data makes the real difference for companies in an ever more complex and non-deterministic business environment. Where credit terms are to be optimized, cross-selling opportunities identified, or early warning signals detected in financial distress, AI-powered credit analysis is what aids in making better decisions for businesses—data-driven decisions that could lead to growth, profitability, and resilience.

The convergence of artificial intelligence and B2B credit analysis heralds a new era of innovation and efficiency in credit risk management. AI will automate more than ever before in processing financial documents, further enhance prediction and insight, and exploit the domain-specific related insights that give AI-driven solutions full power to bring unprecedented smoothness, mitigation, and strategic value realization in operations. The business community is now discovering, rather than just engaging with, the transformative potential of AI to realize truly enhanced competitiveness, increased agility, and far greater resiliency in the ever-cycling markets.

2 Related works

Financial modeling and credit risk management have been undertaken as extensive research in finance, with quite a number of models and methodologies developed for assessing and mitigating risks associated with lending and credit transactions. The section reviews key works and literature that make up a body of knowledge on credit risk modeling, financial analysis, and strategies pertaining to risk management. One of the foundational works in credit risk modeling is the comparison between CreditMetrics and CreditRisk+ by [7]. The paper presents an in-depth comparison between the mathematical structure of the models, highlighting similarities and differences in the measurement of default risk. [7] shows—through functional forms, distributional assumptions, approximation formulas—the finenesses of credit risk assessment. [8] focus much on statistical distributions applied to financial modeling and risk analysis. The works underline major the role of correct representation of uncertainty plays with the choice of statistical distributions. It discusses continuous-time modeling and one of the random processes, the Wiener process, used in mapping changes in the prices of financial assets, bringing out the role of statistical tools in the case. [9] are also concerned with the financial and legal risk for the B2B e-commerce supply chain. They have suggested the construction of risk evaluation systems in order to mitigate these risks by looking into the mechanisms for credit risk generation. The authors offer helpful recommendations on financial risk reduction and control in the B2B electronic-commerce ecosystem since they utilize the KMV model to demonstrate the efficacy of their research on risk evaluation. [10] proposed a novel deep reinforcement learning model for a credit scoring with the objective of maximum correct classification of customers based on the credit risk. In reformulating credit scoring as a sequential classification problem and introducing dynamic reward functions, the authors have shown that their model performs effectively in improving lending decisions. This work contributes to the advancement of credit risk assessment methodologies through innovative machine learning techniques. [1] relate to the application of artificial intelligence in credit risk management in e-commerce platforms. In that respect, the study acknowledges the need for artificial intelligence in credit risk management by predicting credit risks with a high degree of accuracy using machine learning and big data analysis. The authors evidence data-driven decisions and ethical considerations while implementing AI-based credit risk prediction systems that will help in enhancing credit management practices toward better financial outcomes in e-commerce. [11] gives a design and implementation program for B2B supply chain finance platform that integrates blockchain tech with Artificial Intelligence in its credit evaluation and risk control. In other words, the platform aims to make full use of the newest technical media, including blockchain, cloud computing, and AI, to realize efficiency and security within supply chain finance. The work adds to the rapidly growing literature on using technology for enhancing risk management practices in financial transactions. [12], propose a pooling factoring financing strategy with big data credit evaluation technology for SMEs. They shared financing model is such that it reduces financing costs and default risks through the credit evaluation services provided by a B2B platform for SMEs. Thus, with a numerical example, the authors

are able to demonstrate that the approach can extend the financing accessibility to SMEs, hence making this very practical in boosting financial stability and growth. [13] contributed a framework of credit risk prediction with the Optimized FKSVR machine learning classifier. The aim is to enhance predictive accuracy in credit risk assessment and management by optimizing the FKSVR model and integrating it into credit risk management procedures. This framework contributes several ways in which machine learning techniques can be improved for the assessment of credit risk to the literature and hence offers a faster method of risk evaluation to financial institutions. In the paper, [14] examine the economic consequences of risk absorption in indirect auto lending relationships. The authors show how lenders manage risks in B2B markets by assessing the costs and benefits of risk absorption and analyzing dealer responses to the risk-sharing arrangements. This study provides insights into how these are developed vis-à-vis risk management strategies and sustenance of relationships in the context of indirect automobile lending. [15] proposed a methodology for the implementation of credit risk assessment to estimate expected credit losses within the current accounting standard CECL for B2B companies. Expected Loss is computed based on the Probability of Default, Loss Given Default, and Expected at Default. It particularly talks about creditworthiness prediction by developing logistic regression models. This research therefore helps companies to come up with sound credit risk policies by the identification of significant financial ratios and qualitative variables for credit evaluation. [2] presents a structured approach to credit risk evaluation in online supply chain finance using a tailored credit rating index system. It employs a multi-level gray evaluation model based on the Theil index, categorizing creditworthiness into five classes: "excellent," "sub-excellent," "good," "relatively poor," and "poor." The model's feasibility is demonstrated through numerical examples involving corporations, allowing for practical application. By identifying high-risk indices, the approach enables targeted risk control, enhancing banks' decision-making processes. Overall, this methodology offers a valuable framework for assessing credit risk in the evolving landscape of online supply chain finance.

In summary, this literature review portrays the widespread approaches and methodologies in practice within credit risk management, financial modeling, and risk analysis. From some of the traditional models like CreditRisk+, many researchers have moved to the use of innovative techniques with artificial intelligence and machine learning to improvise strategies on risk assessment and mitigation within a financial landscape. These studies are therefore informed by statistical tools, technological developments, and practical frameworks and contribute to the evolving discipline of credit risk management and financial analysis that will yield important insights both for practitioners and researchers.

3 Methodology

After Looking at the multiple approaches stated in the literature which are specific to individual domains, A novel pipeline for generic credit analysis of businesses is proposed in this section.

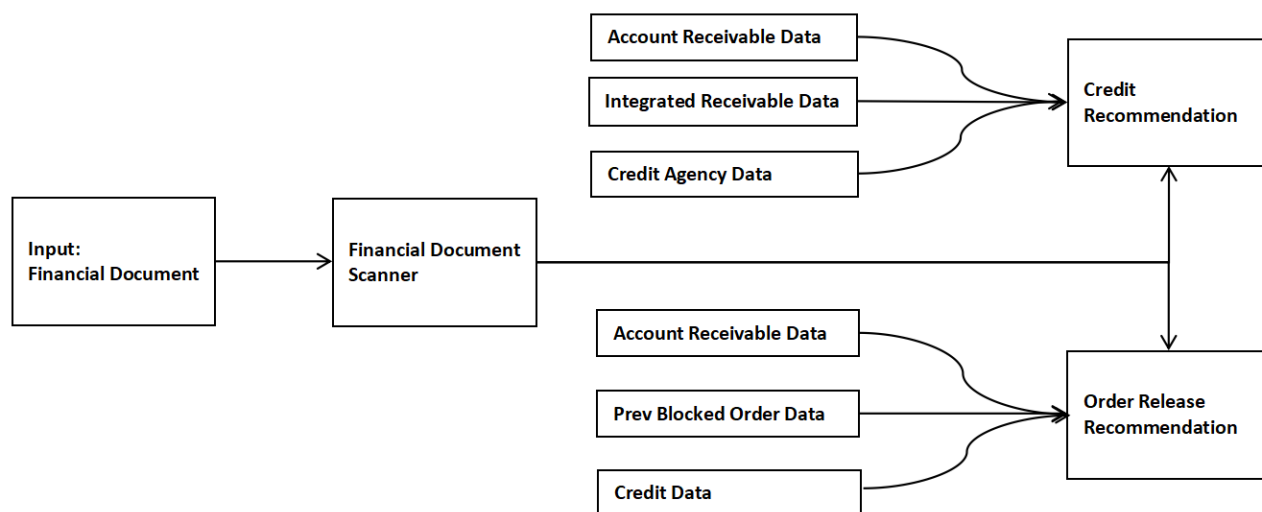


Figure 1: AI based Pipeline for Credit Analysis (Source: Own Results)

3.1 AI-Powered Financial Document Scanning Approach

Understanding financial documents is an important part of any company's credit analysis process which involves extraction of foundational information from financial documents such as profit and loss accounts, balance sheets, and

statements of cash flows. Normally, these documents contain a large amount of data running into several pages, and manual analysis becomes very time-consuming, and relatively subjective. In this scenario, there will be a need for an important, complex pipeline that puts together techniques from artificial intelligence, convolutional neural networks, object detection, and optical character recognition itself to rise to the challenge.

In the Financial Document Scanner use case, the base of the information is built on image data extraction of financial statements received from different companies. These financial statements include all sorts of documents, Balance Sheets, Income Statements, Cash Flow Statements, and others, to represent the financial health of the companies on issues relating to their financial health and performance. Many image preprocessing techniques are applied in order to obtain high-quality images, the enhancements followed in order to improve clarity and contrast, and noise reduction algorithms to remove any kind of distortion or other artefacts. Later, OCR (optical character recognition) technology is used to come up with relaxed textual and numerical information from those preprocessed images, reformatting them into formats that machines could read. Such transformation will open up the route for easy integration of financial data into the analysis pipeline and automate parsing and interpretation of key metrics and indicators of finance.

The financial document scanning pipeline starts by ingesting a PDF document containing a company's financial data. Financial documents can sometimes reach a considerable number of pages, usually from 100 to 150 pages. To treat these documents effectively, the PDF pages are changed into images for further analysis using computer vision algorithms.

The pipeline initiates with CNN-based image classification. The CNN classifiers are trained to find out whether the image contains a table or not, which is of essence in analyzing financial data. This classification stage helps filter only those pages containing tables to be taken for further analysis, therefore cutting down on computational overhead.

First, it detects the pages containing tables, and then it detects the tabular regions in the images. Object detection allows the correct location of a table's position within a document image with complex layouts or noisy backgrounds. The precise demarcation of the tabular regions at this step ensures that only data of relevance is extracted in the subsequent process of OCR.

Identification of tabular regions is followed by cropping the images with respect to the tables and feeding them through OCR algorithms that identify text contained therein. This is a very crucial step through which visual information taken during tables is converted into machine-readable text for further analysis and processing.

Upon successful extraction of text from the tables, the pipeline further classifies tables according to their different types based on the semantic similarity of their content. Semantic classification will then enable the identification of income statements, balance sheets, and cash flow statements that are core building blocks in any financial analysis.

Advanced artificial intelligence techniques in a financial document scanning pipeline have an added advantage over traditional manual methods. Document analysis eats up a substantial amount of time and effort, which gets drastically reduced as it gets automated, freeing the analyst to work on the interpretation of information and decision-making. In addition, the accuracy and consistency of AI-based approaches reduce errors in manual data extraction procedures.

The scalability of this pipeline that makes it capable of handling a large volume of financial documents, which applies to organizations working with varied portfolios or frequent data updating. Moreover, the ability to adjust to documents in various layouts and formats makes the pipeline even more scalable and authoritative, crossing organizational and domain barriers.

Despite the advancements offered by AI-driven document scanning, Certain things have to be taken into consideration for AI-driven document scanning to be at once effective and reliable. First, invariance to the quality of documents introduces noise, blur, or irregular formatting factors which may impact the accuracy of the extraction process. Further, the pipeline will require continuous monitoring and improvement in accordance with the changing business requirements and alignment with the standards and regulations of the industry. The financial document scanning pipeline is, therefore, a break-through innovation on credit analytics that places automation and artificial intelligence at the core of extracting key information from financial documents.

In conclusion, The Financial document scanner makes it possible to process, sort, and understand complex financial data. It helps people make smart choices that can lower risk in a company. The system allows for the handling, categorizing, and making sense of tricky money-related details. This leads to insights that help make well-informed choices, which can cut down on risk in a business. In other words, as AI keeps getting better fast, the ways of document scanning will keep changing and getting better. This will lead to ongoing improvements in how humans look at finances and make choices.

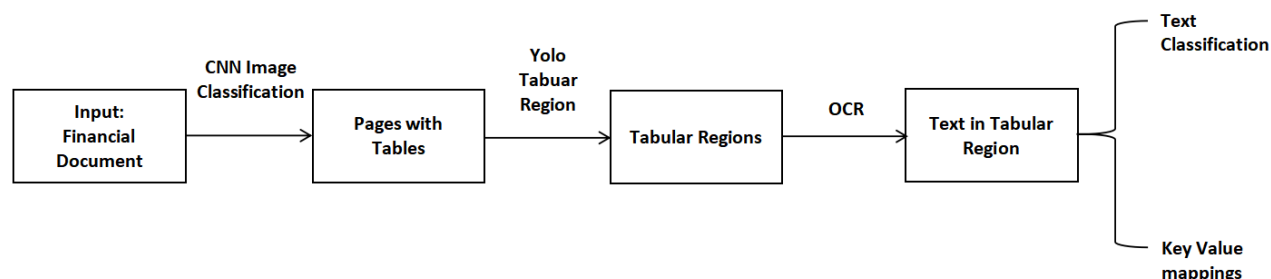


Figure 2: Financial Document Scanner Pipeline (Source: Own Results)

3.2 AI-Powered Credit Limit Recommendation

Optimizing the credit limit for each buyer will bring out an ideal balance between credit use and financial risk exposure. Appropriate setting can free huge capitals running into millions of dollars and at least reduce the provision for credit loss on any financial statement by a certain percentage. A balance sheet element that companies set aside to account for expected losses from delinquent accounts is the bad debt allocation or provision for credit losses. This includes the operating expenses or loss one could incur from sales that are not collectible.

In an unpredictable economic landscape, a consistent review of buyers credit health has to be enhanced by the suppliers. The process for such evaluation is currently manual and only reactive to extend credit monitoring and review. The new AI-driven credit recommendation feature, however, is designed to make this a proactive, real-time credit review. Drawing from Integrated Receivables, Accounts Receivable data, and Credit Agency data, the AI model can recommend to the user credit extensions, upgrades, or downgrades for each buyer. These recommendations include specific values and ranges, therefore providing an all-round solution for credit limit optimization. In case of an upgrade in credit limit predicted by the AI model, it would recommend increasing their credit limit with options to accept or reject by the analysts. On the other hand, in the case of a downgrade, it would recommend that the credit limit be decreased with again options to either accept or reject by the analysts. For credit extensions, this model recommends holding at the current credit limit but extending for some period, where analysts have an option to either accept or reject. Following are the key business benefits of implementing an AI-powered credit recommendation system:

- **Automated Proactive Credit Limit Optimization:** Streamlines the process of assessing and adjusting credit limits for buyers, ensuring timely and appropriate credit allocations,
- **Fast-Track Compliance:** Enhances compliance by automating and standardizing credit review processes,
- **Increased Credit Limit Assessments:** Boosts the percentage of buyer credit limit assessments conducted each week,
- **Time Savings:** Reduces the time and effort by automating the review process.
- **Unlocks Sales Opportunities:** Reduces blocked orders, facilitating smoother sales operations,
- **Automation for Low-Medium Credit Limit Buckets:** Enables automated reviews for customers with low to medium credit limits,
- **Reduces Bad Debt:** Helps decrease bad debt from AR and increases the overall accounts receivable.

The Credit Limit Recommendation use case dataset offers an extensive collection of information gathered from a number of channels to project a composite view about customer creditworthiness and payment behavior. The essential components of the dataset are receivables data, financial data such as EBITDA, credit agency data comprising the credit score of a company from various credit agencies, invoice data like details on transactions constituting the amount of the invoice, date due, and whether it is paid or not, etc., which form elementary inputs for assessing customer credit risk. I also consider previous payment history and the credit history of customers as important components in establishing patterns and trends that may signify creditworthiness and reliability. To this respect, I try to make an analysis of the past payment behavior under very detailed parameters: timeliness of payments, frequency of late payments, and adherence to credit terms, each of these gives us valuable inputs into the credit risk profiling of individual customers.

3.3 AI-Powered Recommendation System for Efficient Order Release

In the realm of B2B transactions, managing blocked orders is a significant challenge. The critical decision point is not whether to release these orders but rather when to do so. Credit teams typically consider several key attributes in their

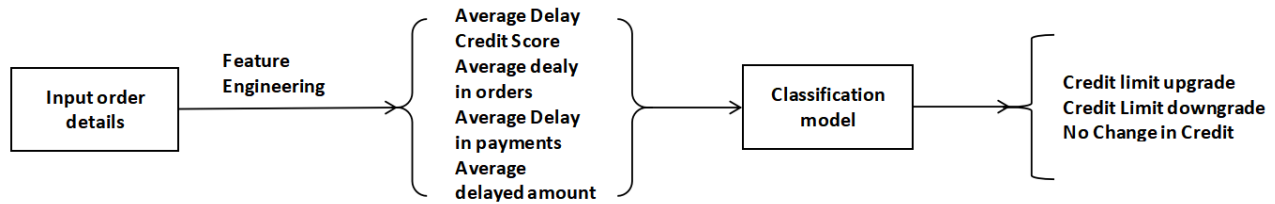


Figure 3: Credit Limit Recommendation Engine Pipeline (Source: Own Results)

decision-making process, including the risk profile of the customer, past payment behavior, and instances of credit limit overshoot. These decisions are governed by global credit policies and are consistently reflected in the transactional data.

In this respect, data used by analysts to arrive at such decisions can also be used in the training of AI models. These models can mimic the strategies that credit analysts use in making decisions to recommend immediate release of orders or other decisions, temporary or permanent, where a customer's credit limit needs to be increased or decreased. In addition, these AI-driven recommendations can help in efficiently automating the process of releasing blocked orders.

The AI-powered recommendation system supports the credit analysts with exact recommendations on unblocking a blocked order. It takes into consideration various accompanying actions, such as taking collateral or increasing the credit limit, and gives the reasoning for these recommendations. It simulates the decision-making patterns of credit analysts based on past and real-time information; hence, the recommendations do not, at any time, divert from the predefined global credit policies.

The implementation of an AI-driven recommendation system offers several significant business benefits:

- **Increase in Analyst Productivity:** By accelerating the decision-making process for blocked orders, analysts can handle more cases in less time, leading to a more efficient workflow.
- **Sales Acceleration:** Faster decision-making on blocked orders translates to quicker order releases, thereby speeding up sales cycles and revenue realization.
- **Minimized Business Disruption:** Timely release of blocked orders minimizes the impact on the business operations of buyers, maintaining healthy business relationships and ensuring smooth transaction flows.

The major business metrics the AI-powered recommendation system would impact is the average time taken to release a blocked order. To that effect, this can be significantly reduced using the system in smoothing and partially automating the process of decision-making, hence improving operational efficiency and customer satisfaction. It uses comprehensive input data to predict accurate recommendations that includes Receivables data: This includes invoice date, due date, clearing date, payment terms etc. Order data consisting of order date, order created date, order current status etc. Customer Data regarding Credit Limit, Credit Exposure etc. The fourth category is, the credit limit history data: Previous credit limit, Updated credit limit, reason for changes etc.

The actual efficiency of blocked order prediction lies in the quality and comprehensiveness of data input into the AI model. The data comes with various dimensions on the aspects of AR, Order Processing, customer-profiling data, promise-to-pay, and credit limit history. This integration of multifaceted data makes the model usefully accurate in its predictions with respect to blocked orders. One of the important data about the financial transactions taking place between a company and its customers is the Account Receivables data. It uses the following attributes: Due Date, Clearing date, Invoice Date in date type formats; Invoice Amount, Payment Terms, Order ID, Disputed Amount in integer formats.

4 Results

The implementation of the AI-powered financial document scanning model yielded substantial business benefits across various dimensions:

- **Efficiency in Document Analysis:** By automating the extraction and analysis of financial data from extensive documents, the model significantly reduced the time and effort required compared to manual methods. This efficiency allowed analysts to focus more on interpretation and decision-making rather than data entry.

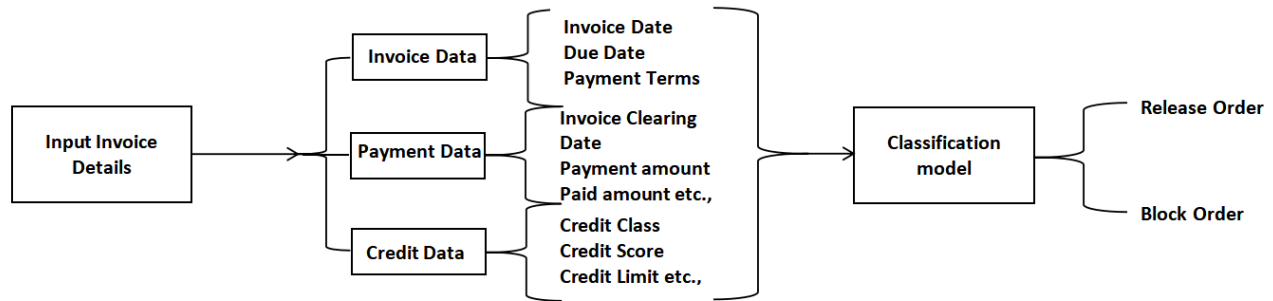


Figure 4: Order Release Recommendation Engine Pipeline (Source: Own Results)

- **Improved Accuracy and Consistency:** Leveraging AI-driven approaches minimized errors associated with manual data extraction, ensuring more reliable financial insights. The consistency in data extraction contributed to more robust financial analysis and reporting.
- **Scalability:** The model demonstrated scalability by efficiently processing large volumes of financial documents. This capability was particularly beneficial for organizations with diverse portfolios or frequent updates, enabling them to handle substantial data influxes without compromising on analysis quality.
- **Enhanced Decision-Making:** Providing timely and accurate financial information enabled quicker and more informed decisions by analysts and stakeholders. This enhancement in decision-making speed was crucial in adapting to market changes and seizing business opportunities promptly.
- **Adaptability:** The model's versatility across various document formats and layouts accommodated different industries and business needs. This adaptability ensured that the AI-powered system could be seamlessly integrated into existing workflows across diverse organizational settings.
- **Risk Mitigation and Compliance:** The comprehensive and consistent analysis of financial data facilitated by the model helped in identifying and mitigating risks effectively. Furthermore, by ensuring accurate data extraction and analysis, the model supported adherence to industry regulations and standards, thereby enhancing overall compliance.
- **Operational Cost Savings:** By automating manual data entry and analysis processes, the model reduced operational costs significantly. This cost-saving measure allowed organizations to allocate resources more efficiently towards higher-value tasks and strategic initiatives.
- **Continuous Improvement and Competitive Advantage:** The model facilitated ongoing refinement and enhancement of the scanning pipeline to meet evolving business requirements and technological advancements. By leveraging AI for advanced financial analysis, organizations gained a technological edge, positioning them ahead in the competitive landscape.

5 Conclusion

Artificial intelligence has been integrated into the analysis of financial documents, credit recommendation, and order release management to give a seamless advancement in the space of credit analysis and business operations. In particular, the financial document scanning pipeline with the help of Convolutional Neural Networks for object detection and Optical Character Recognition has taken a different turn in the extraction and processing of resultant complex financial data. This has substantially increased efficiency, accuracy, and reliability by allowing analysts to spend their time on interpretive work rather than manual extraction.

The AI-powered credit recommendation system has automated the typically manual and reactive credit limit optimization process. It gives real-time, accurate credit limit recommendations by leveraging the three comprehensive sources of data: Integrated Receivables, AR, and Credit Agency data. This has come to aid in smoothing credit assessments, enhancing compliance, reducing bad debt, and unlocking new sales opportunities, hence contributing to the overall financial health of businesses.

The AI-powered recommendation system for blocked order management brought a significant performance increase in decision-making and additional productivity for analysts to help drive faster sales cycles. The system, while serving up

Table 1: Overview of Relevant Works

Authors	Approach
Gordy et al. [7]	Comparison between CreditMetrics and CreditRisk+, focusing on the mathematical structures of the models, and measurement of default risk.
Kliesik [8]	Focus on statistical distributions applied to financial modeling and risk analysis, discussing continuous-time modeling and the Wiener process
Liu et al [9]	Construction of risk evaluation systems for financial and legal risk in the B2B e-commerce supply chain, using the KMV model.
Paul et al [10]	Deep reinforcement learning model for credit scoring, reformulating credit scoring as a sequential classification problem with dynamic reward functions.
Lal et al. [1]	Application of artificial intelligence in credit risk management on e-commerce platforms, predicting credit risks with machine learning and big data analysis.
Xue [11]	Design and implementation of a B2B supply chain finance platform integrating blockchain and AI for credit evaluation and risk control.
Zhang and Liu [12]	Pooling factoring financing strategy with big data credit evaluation technology for SMEs, demonstrated with a numerical example.
Usha Devi [13]	Framework of credit risk prediction with the Optimized FKSVR machine learning classifier, enhancing predictive accuracy in credit risk assessment and management.
Zhou et al. [14]	Examination of the economic consequences of risk absorption in indirect auto lending relationships, analyzing dealer responses to risk-sharing arrangements
Abano [15]	Methodology for implementing credit risk assessment to estimate expected credit losses under the current accounting standard CECL for B2B companies, using logistic regression models to predict creditworthiness.
Gummadi Sai Dheeraj	Enhancing Credit Decision with AI: A Pipeline for Automated Financial Document Analysis And Creditworthiness Prediction

automated releases for blocked orders and delivering data-driven recommendations, minimizes business disruptions and maintains healthy buyer relationships, ensuring smooth transaction flows.

These AI-driven solutions have been demonstrated the capability to handle large amounts of data, flexibility to adapt to document formats and layouts, and produce consistent results with high accuracy. All the advantages of automation includes, time savings, reduction of errors, and improvement in decision-making underscore the disruptive potential that AI has for financial analysis and business operations.

The way AI is growing definitely gives hope for many more innovations to come, Coupled with the continual refinement and adaptation of such AI models to evolving business requirements and industry standards is important for their sustained effectiveness and reliability. The advances described in this paper couple the potential of AI to drive large, never-before improvements in credit analysis and operational efficiencies, hence empowering an organization to make contextually informed decisions while mitigating risks.

References

- [1] Santosh Lal, Manjula Pattnaik, G Srilakshmi, Seema Singh, and Archana Ravindra. Artificial intelligence based risk management of credit sales in e-commerce.
- [2] Jinzhao Shi, Ju'e Guo, Shubin Wang, and Zhanhao Wang. Credit risk evaluation of online supply chain finance based on third-party b2b e-commerce platform: an exploratory research based on china's practice. *International Journal of u-and e-Service, Science and Technology*, 8(5):93–104, 2015.
- [3] Diana Wong. B2b partnerships: creating win-win situations. 2016.
- [4] Olga Rogacka. The b2b sales meaning, examples, and strategies. 2024.
- [5] Team Happay. Order to cash (o2c) process: A step-by-step approach. 2023.
- [6] The Investopedia Team. What is credit analysis? how it works with evaluating risk. 2022.
- [7] M.B. Gordy, Federal Reserve System. Division of Research, and Statistics. *A Comparative Anatomy of Credit Risk Models*. Finance and economics discussion series / Federal Reserve System. Federal Reserve Board, 1998.

- [8] Daniel Buc-Tomáš Klieštík. Aspects of statistics in terms of financial modelling and risk. 2013.
- [9] Cheng-yong Liu, Tian-yu Dong, and Ling-xing Meng. The prevention of financial legal risks of b2b e-commerce supply chain. *Wireless Communications and Mobile Computing*, 2022(1):6154011, 2022.
- [10] Sudipta Paul, Agam Gupta, Arpan Kumar Kar, and Vinay Singh. An automatic deep reinforcement learning based credit scoring model using deep-q network for classification of customer credit requests. In *2023 IEEE International Symposium on Technology and Society (ISTAS)*, pages 1–8. IEEE, 2023.
- [11] Mengdan Xue. Design and implementation of a b2b e-commerce supply chain finance platform based on blockchain and artificial intelligence technology.
- [12] Jie Zhang and Yuehui Liu. Pooling factoring financing strategy based on the big data credit evaluation technology of b2b platform. *Electronic Commerce Research*, pages 1–17, 2023.
- [13] Dr Neera Batra Usha Devi. A framework for credit risk prediction using the optimized-fksvr machine learning classifier.
- [14] Qiang Kris Zhou, Huanhuan Shi, Shrihari Sridhar, and Adithya Pattabhiramaiah. The economic consequences of risk-absorption in b2b relationships: Evidence from indirect auto lending. *Huanhuan and Sridhar, Shrihari and Pattabhiramaiah, Adithya, The Economic Consequences of Risk-Absorption in B2B Relationships: Evidence from Indirect Auto Lending (January 10, 2024)*, 2024.
- [15] Karina Couto Xavier Abano. B2b credit risk management. 2021.